

SV16 FPGA Board

Component placement redesign and portable project handoff | 8 October 2026

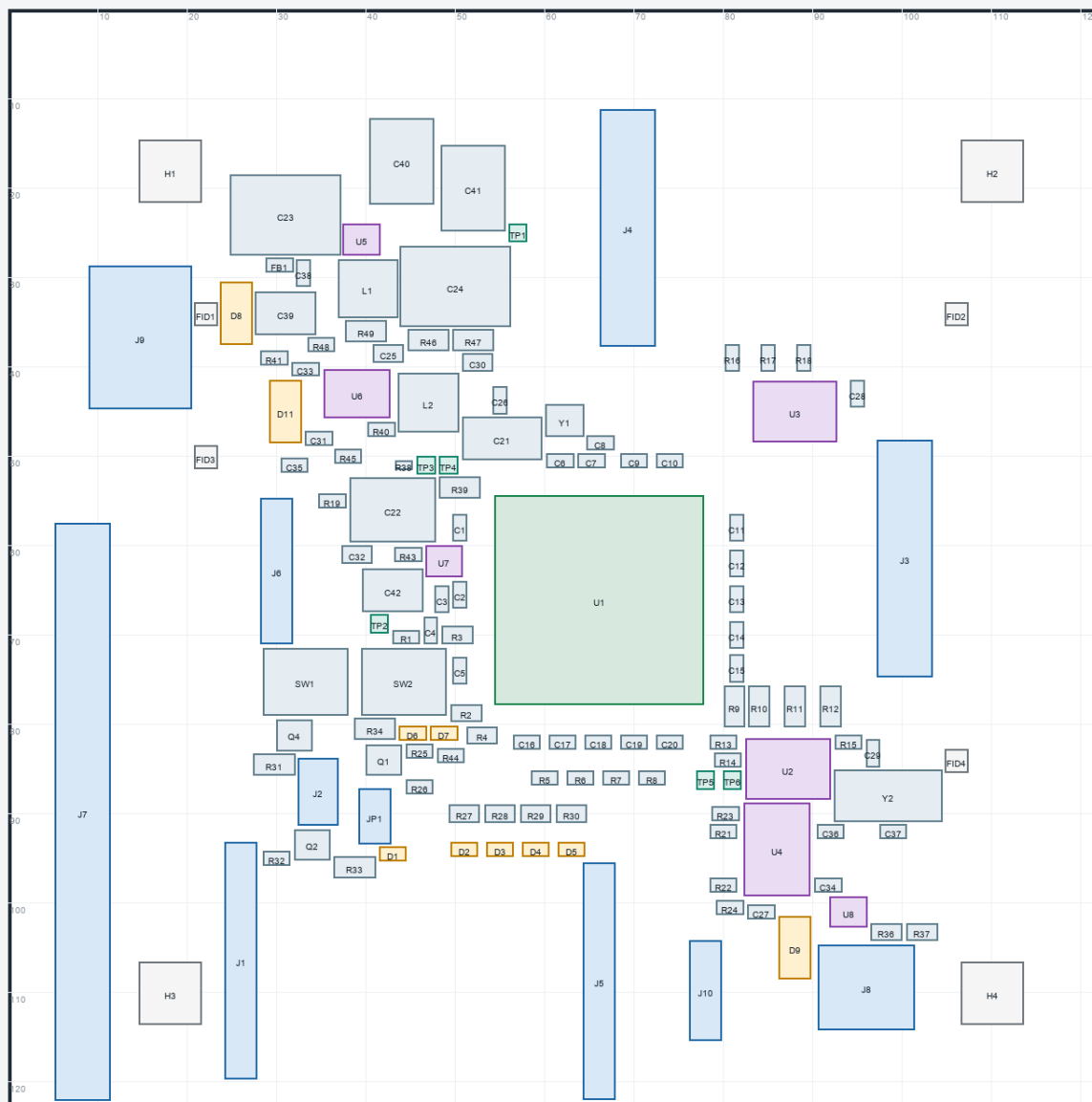
Placement stage complete

141 footprints arranged across the 122.0 x 122.75 mm board outline.

Routing still open

This new placement has no tracks, vias, or copper zones yet.

SV16 • routing-oriented component placement



Placement preview uses each footprint courtyard to show package spacing. Blue: connectors; purple: ICs; green: FPGA/test points; amber: diodes; gray-blue: passives.

1. Placement changes and verification

The native board was reorganized as a placement-only baseline. Electrical pad numbers and net assignments were left intact; the existing tracks from the older routing experiment were not carried into this new layout.

Placement decisions

- **FPGA:** kept centered to preserve breakout space around all four 0.5 mm-pitch sides.
- **Expansion headers:** J4 (B signals) north of the FPGA, J3 (A8-A15 and SPI) east, and J5 (A0-A7) south.
- **Power:** barrel input, switching regulators, inductors, bulk capacitors, protection diodes, and rail test points grouped in the upper-left area.
- **Programming and status:** JTAG, reset/configuration switches, transistor indicators, and LEDs grouped to the left and south-west.
- **USB/UART:** USB-C, ESD protection, CH340, its clock, series parts, and console header grouped in the south-east.
- **Mechanical and assembly:** four mounting holes and four fiducials kept at their original coordinates; all footprint courtyards have clearance.

Measured checks

Check	Result	Meaning
Footprints / pads	141 / 591	Same footprint and pad population as source.
Pad-to-net assignments	Exact match	All 591 pad net names and pad numbers were verified unchanged.
Courtyard overlaps	0	No footprint body/courtyard overlaps.
Silkscreen collisions	0	No DRC silk overlaps or silk-to-copper collisions.
Copper objects	0 tracks, 0 vias, 0 zones	The file is not electrically routed.
Unconnected items	353	Expected for a board with no routed copper.
KiCad library warnings	141 warnings	The temporary DRC environment has no standard footprint library table. These are library lookup warnings, not geometric clearance errors.

Important: the four-layer stackup is configured, but the current board has not yet been routed or passed full connectivity DRC.

2. Files and portability

Keep the following files together when moving the project to another computer. The current native placement board is the source for a fresh routing pass.

Role	Path
Current placement	hardware/sv16_board/sv16_board_placed_native.kicad_pcb
KiCad project settings	hardware/sv16_board/sv16_board_placed_native.kicad_pro
Pre-change backup	hardware/sv16_board/placement_artifacts/sv16_board_original_preplacement.kicad_pcb
Placement preview	hardware/sv16_board/placement_artifacts/relayout_preview.png
Reusable placement scripts	scripts/relayout_sv16_native.py; scripts/sv16_place_native.py; scripts/place_sv16_reference_fields.py; scripts/render_sv16_placement.py
Project notes	hardware/README.md

Four-layer plan

Layer	Intended role
F.Cu	Signals and FPGA fanout.
In1.Cu	Ground plane.
In2.Cu	3V3 plane with isolated power-rail routes.
B.Cu	Signals and ground copper.

The old files ``routing_state.json``, ``routing.kicad_pcb.txt``, and ``sv16_board_working.kicad_pcb`` describe the previous component positions. Do not resume or merge that copper into the new placement without re-routing and rechecking it.

Router environment

The route script uses KiCad Python (``pcbnew``), NumPy, and Shapely 2.2.0. The current host lacks Shapely, so a new autoroute was not run. The router now accepts separate ``--state`` and ``--snippet`` paths to keep future checkpoints paired with this board.

3. Remaining engineering work

The earlier routed draft had 76 of 112 signal nets joined (68%), but that measurement belongs to the old placement. For the current board, routing is 0/112 until copper is generated from this layout.

- **Route fresh:** fan out the FPGA, route signals and power islands across the four copper layers, and pour the GND/3V3 planes. Keep the new route state and snippet in separate files.
- **Integrate and verify:** load the generated copper into a named working board, refill zones in KiCad, compare connectivity to the schematic, and check for shorts, clearance errors, dangling vias, and board-edge violations.
- **Resolve open connections:** require all 112 signal nets and all power islands to connect, with zero unconnected items before fabrication output.
- **Review BOM fit:** confirm the inductor values and current ratings, switching-regulator passives, and the purchased 25 MHz active oscillator footprint/pinout.
- **Release only after signoff:** run DRC in a KiCad installation with standard libraries enabled, then generate Gerbers, drill files, BOM, and placement data.

Known BOM check

The cart records both 10 uH and 33 uH inductors while the current PCB assigns 10 uH to L1 and L2. Confirm the intended L2 value against the MP1584 design and the purchased inductor current rating. Confirm the active oscillator package/pinout at Y1 before release.

Placement: complete and verified.	Full PCB: in progress; not ready to manufacture.
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Current authoritative board: hardware/sv16_board/sv16_board_placed_native.kicad_pcb